Automated Wheatgrass Production System – Phase II (Digital Twin) Hardware Maintenance Manual

Client Organization: Tardigrade Ltd. and the Foundation of the Energy Collective

Client Name: Sebastien Chabot

Team PB-35

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Authors: Tahsin Hasan, Hamza Islam, Sam Pan, Ao Sun

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1. Document Purpose and Scope

This maintenance manual covers the deployed monitoring hardware that supports the wheatgrass digital twin. It focuses on operation support, preventive maintenance, fault isolation, recovery, and replacement actions for the installed hardware stack. The manual does not cover mechanical redesign of the conveyor or advanced stretch-goal features that were outside the Phase II core scope.

- ESP32 sensing and communication node(s)
- BME280 temperature/humidity sensor module(s)
- TSL2591 light sensor module(s)
- YF-S201 water-flow sensor, where installed
- Raspberry Pi backend hosting Home Assistant and API-related services
- Portable Wi-Fi or local network equipment used to connect ESP32 node(s) to the backend
- Power adapters, SD card storage, and wiring associated with the monitoring subsystem

2. System Overview

The hardware subsystem collects environmental and operational data from the wheatgrass machine and forwards it to the Raspberry Pi backend for monitoring, visualization, and logging. The design uses an ESP32 microcontroller, digital sensors, Wi-Fi networking, and a Raspberry Pi running Home Assistant and API services.

Layer	**Main Components**	**Role in Operation**	**Maintenance Relevance**
Sensing	BME280, TSL2591, YF-S201	Measure temperature, humidity, light, and water flow	Clean, inspect, verify signal quality, and replace failed sensors
Control	ESP32 dev board	Reads sensors and transmits data over Wi-Fi	Check power, boot state, reconnection behavior, and firmware configuration
Network	Portable Wi-Fi / local LAN path	Provides transport between ESP32 and Raspberry Pi	Check signal strength, IP connectivity, and broker reachability
Backend	Raspberry Pi, Home Assistant, storage media	Receives, stores, and visualizes data	Back up configuration, check logs, maintain SD card health, and verify service uptime

Table 1. System Functions.

3. Maintainer Roles and Responsibilities

Role	Primary Responsibilities	Typical Triggers
Operator / Technician	Perform visual inspection, verify dashboard status, clean exposed sensors, and report abnormalities.	Daily checks, alert response, visible hardware issues.
System Maintainer	Diagnose ESP32, network, sensor, or power issues; perform component replacement; verify end-to-end recovery.	Repeated offline status, unstable readings, failed power-cycle recovery.
Backend Maintainer	Maintain Raspberry Pi, backups, SD image integrity, Home Assistant services, and restore procedure.	Storage issues, corrupted configuration, service crash, OS rebuild.

Table 2. Maintainer roles and corresponding responsibilities

4. Safety and Handling Precautions

- Disconnect power before rewiring sensors, replacing modules, or moving the ESP32 or Raspberry Pi enclosure.
- Keep the microcontroller, power supply, and exposed electrical connections away from mist, standing water, and irrigation spray zones.
- Use only regulated power adapters of the intended rating. For the ESP32 subsystem, the project context references 5V/2A adapters during prototyping and deployment support.
- Avoid pulling sensor leads by the wire. Hold connector housings directly when disconnecting modules.
- Use anti-static precautions when handling the Raspberry Pi, ESP32 board, and sensor breakout boards.
- After maintenance, confirm cable routing does not interfere with trays, conveyors, sprinklers, or access doors.

5. Routine Maintenance Schedule

Interval	Task	Acceptance Criteria	Record to Keep
Daily/Before operation	Check dashboard for sensor freshness, node online status, and abnormal alerts.	All expected sensors update normally; node status is online; no persistent unexplained alert.	Operator log entry and screenshots if abnormal.
Weekly	Inspect sensor placement, cable strain relief, connector fit, and signs of moisture ingress or contamination.	No loose wiring, no corrosion, no cracked housings, no direct spray exposure to electronics.	Inspection checklist.
Monthly	Clean sensor surfaces and nearby mounting area; verify power adapters are not overheating; confirm Raspberry Pi free storage and recent backup.	Sensor openings unobstructed; adapters warm but not excessive; backup completed successfully.	Maintenance log with date and initials.
Quarterly	Perform functional verification: power-cycle ESP32, confirm reconnection, confirm valid readings change with environment, and verify backend data logging.	System returns to service automatically; readings are plausible; backend history remains continuous.	Functional test record.
After incident	Inspect and test after any power outage, water leak, network change, or physical relocation.	No latent faults remain; data path and alerts function again.	Corrective-action report.

Table 3. Routine check-up schedule

6. Detailed Maintenance Procedures

6.1 Visual Inspection

- Confirm the ESP32 and Raspberry Pi are mounted securely and protected from moisture exposure.
- Look for damaged insulation, bent pins, corrosion, discoloration, loose Dupont leads, and cracked sensor boards.
- Confirm the light sensor is not blocked by dust, tape, or plant material.
- Confirm the BME280 air path is not sealed by condensation, debris, or accidental enclosure changes.
- If a water-flow sensor is installed, check for leaks, clogged lines, and secure hose routing.

6.2 Cleaning

- Power down the local sensing node before cleaning around exposed electronics.
- Use a dry lint-free cloth or soft brush for dust removal. Do not soak sensor openings or use compressed air at very high pressure.
- Do not apply liquid cleaner directly to the ESP32, BME280, Raspberry Pi, or SD card contacts.

6.3 Power and Wiring Check

- Verify the correct adapter is connected and cable jackets are intact.
- Check that the ESP32 powers up consistently and that the Raspberry Pi boots without undervoltage or repeated restart symptoms.
- Inspect all I2C and digital signal wiring for secure termination and adequate strain relief.
- Replace cables or connectors that intermittently disconnect when lightly moved.

6.4 Backend Backup and Storage Check

- Back up the Raspberry Pi system image or key Home Assistant and broker configuration before major changes.
- Retain at least one known-good spare microSD card or restorable image for rapid recovery.
- Check for database/log growth that may exhaust storage and degrade performance.
- Document backup date, image version, and restore location.

7. Startup and Recovery Verification

- Apply power to the Raspberry Pi backend and allow core services to start fully before judging the sensing node status.
- Apply power to the ESP32 node and confirm sensor values begin updating on the backend dashboard.
- If the network was unavailable at boot, verify that the ESP32 attempts reconnection automatically rather than remaining permanently offline.
- After a power-cycle test, confirm the first valid payload or dashboard update returns and that readings are plausible for the environment.

8. Troubleshooting Guide

Symptom	Likely Cause	Immediate Action	Corrective Action
ESP32 node offline on dashboard	Power loss, Wi-Fi failure, boot failure, or configuration issue	Check power LED/boot, verify hotspot or LAN availability, and reboot node once	Inspect adapter and wiring; reflash or restore firmware configuration if repeated failure continues
Temperature / humidity reading stuck or implausible	Sensor contamination, disconnected BME280, or failed I2C link	Inspect connector seating and compare with room conditions	Clean or replace BME280; verify I2C wiring and address configuration
Light value always very low or unchanged	Obstructed TSL2591, wiring issue, or damaged sensor	Remove obstruction and test under different light levels	Replace sensor if no response change occurs
No water-flow reading during irrigation	Flow sensor blockage, reversed installation, wiring fault, or no actual flow	Check for active watering and inspect plumbing path	Clear blockage or replace YF-S201; verify pulse input wiring
Backend not logging new data	Raspberry Pi service issue, broker problem, storage fault, or network break	Check Home Assistant/service status and free disk space	Restart affected service; restore from backup or replace SD card if corruption is detected
Frequent disconnect / reconnect behavior	Weak Wi-Fi signal, power instability, or overloaded adapter	Relocate node/router temporarily and inspect adapter temperature	Improve network placement, replace adapter, or reduce cable/connector instability

Table 4. Hardware potential issues, causes and actions

9. Component Replacement Procedures

9.1 Replacing a Sensor Module

- Power down the sensing node.
- Label the old connection before removal.
- Install the new sensor module with the same connector orientation and secure mounting position.
- Power on and verify the sensor is detected and the value is plausible.
- Record replacement date, part number, and post-replacement test result.

9.2 Replacing the ESP32 Node

- Back up or export the current ESPHome / firmware configuration if available.
- Move sensor connections one by one or document the full wiring map first.
- Load the approved firmware/configuration onto the replacement board.
- Verify Wi-Fi credentials, topic/API settings, and backend reachability.
- Run a power-cycle and end-to-end data check before returning the system to service.

9.3 Replacing Raspberry Pi Storage or Rebuilding the Backend

- Prepare a new imaged microSD card or restore from the latest validated backup.
- Boot the Raspberry Pi and confirm network address, Home Assistant availability, and broker functionality.
- Reconnect the sensing node and verify that fresh data appears again.
- Preserve the failed storage card for post-failure analysis where practical.

10. Acceptance Criteria After Maintenance

- All expected sensors report values that are numeric, current, and plausible for the environment.
- The microcontroller node appears online and resumes transmission after a controlled restart.
- The Raspberry Pi backend stores and displays fresh data without repeated service errors.
- No exposed electronics remain in a location vulnerable to direct water or sustained condensation.
- The maintenance action, outcome, and any changed configuration are documented.